

Masterclass in Python

Exercise Set 2: Looping Constructs

30 Exercises on Single-Level Loops, Nested Loops, and Loops with Branching

Instructions

This exercise set contains 30 programming problems focused on looping constructs in Python. The exercises are organized into three sections:

Section A (Q1-Q10): Single-Level Looping - Basic for and while loops

Section B (Q11-Q20): Nested Looping - Loops within loops

Section C (Q21-Q30): Nested Loops with Branching - Complex control flow

For each exercise, write your Python code. Use lists and dictionaries as needed.

Section A: Single-Level Looping

These exercises cover basic for and while loops. Focus on understanding loop iteration, loop variables, and accumulator patterns.

Q1. Sum of First N Natural Numbers

Write a program that takes a positive integer N from the user and calculates the sum of all natural numbers from 1 to N using a for loop. Display the result.

Example: If N = 5, output should be 15 (1+2+3+4+5)

Q2. Count Vowels in a String

Write a program that takes a string from the user and counts how many vowels (a, e, i, o, u - both uppercase and lowercase) are present in it. Use a for loop to iterate through each character.

Example: Input: "Hello World", Output: 3

Q3. Multiplication Table

Write a program that takes a number from the user and displays its multiplication table from 1 to 10. Each line should show the complete multiplication expression.

Example: For input 7: "7 x 1 = 7", "7 x 2 = 14", ... "7 x 10 = 70"

Q4. Reverse a List

Given the list numbers = [10, 20, 30, 40, 50], write a program to create a new list containing the elements in reverse order using a for loop. Do not use the built-in reverse() method or slicing.

Expected Output: [50, 40, 30, 20, 10]

Q5. Factorial Using While Loop

Write a program that takes a non-negative integer from the user and calculates its factorial using a while loop. Remember: $0! = 1$ and $n! = n \times (n-1) \times (n-2) \times \dots \times 1$

Example: Input: 5, Output: 120

Q6. Find Maximum in a List

Given the list values = [34, 12, 89, 23, 67, 45, 90, 56], write a program to find the maximum value using a for loop. Do not use the built-in max() function.

Expected Output: 90

Q7. Sum of Dictionary Values

Given the dictionary prices = {"apple": 50, "banana": 30, "orange": 40, "mango": 80}, write a program to calculate the total of all prices using a for loop.

Expected Output: 200

Q8. Print Pattern - Ascending Stars

Write a program that takes a number N from the user and prints a pattern of stars. The first line should have 1 star, second line 2 stars, and so on until N stars.

Example for N=4:

```
*  
**  
***  
****
```

Q9. Count Digits in a Number

Write a program that takes a positive integer from the user and counts how many digits it has using a while loop. Do not convert the number to a string.

Example: Input: 12345, Output: 5

Q10. Filter Even Numbers from List

Given the list data = [3, 8, 15, 22, 7, 36, 41, 50, 19, 64], write a program to create a new list containing only the even numbers using a for loop.

Expected Output: [8, 22, 36, 50, 64]

Section B: Nested Looping

These exercises require loops within loops. Pay attention to how the inner loop executes completely for each iteration of the outer loop.

Q11. Multiplication Table Grid

Write a program to display a 5×5 multiplication table grid. The entry at row i and column j should be $i \times j$. Format the output neatly with proper spacing.

Expected format (first 3 rows shown):

```
1  2  3  4  5
2  4  6  8 10
3  6  9 12 15
```

Q12. Right-Angled Triangle Pattern

Write a program that takes N from the user and prints a right-angled triangle pattern using numbers.

Example for $N=4$:

```
1
1 2
1 2 3
1 2 3 4
```

Q13. Print All Pairs from Two Lists

Given `list1 = [1, 2, 3]` and `list2 = ['a', 'b', 'c']`, write a program to print all possible pairs where the first element comes from `list1` and the second from `list2`.

Expected Output: (1, 'a'), (1, 'b'), (1, 'c'), (2, 'a'), (2, 'b'), (2, 'c'), (3, 'a'), (3, 'b'), (3, 'c')

Q14. Inverted Number Triangle

Write a program that takes N from the user and prints an inverted triangle pattern.

Example for $N=5$:

```
5 4 3 2 1
5 4 3 2
5 4 3
5 4
5
```

Q15. Matrix Row Sums

Given the 2D list (matrix):

```
matrix = [[1, 2, 3], [4, 5, 6], [7, 8, 9]]
```

Write a program to calculate and display the sum of each row using nested loops.

Expected Output: Row 1 sum: 6, Row 2 sum: 15, Row 3 sum: 21

Q16. Hollow Square Pattern

Write a program that takes N from the user and prints a hollow square pattern of size N×N using asterisks.

Example for N=5:

```
* * * * *
*       *
*       *
*       *
*       *
* * * * *
```

Q17. Flatten a 2D List

Given the nested list:

```
nested = [[1, 2], [3, 4, 5], [6], [7, 8, 9, 10]]
```

Write a program to create a single flat list containing all elements using nested loops.

Expected Output: [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]

Q18. Character Frequency in Word List

Given the list words = ["apple", "banana", "cherry"], write a program to count and display how many times each character appears across all words. Use a dictionary to store the counts.

Sample Output: {'a': 5, 'p': 2, 'l': 1, 'e': 2, ...}

Q19. Diamond Pattern (Upper Half)

Write a program that takes N from the user and prints the upper half of a diamond pattern (pyramid centered).

Example for N=4:

```
  *
 ***
*****
*****
```

Q20. Dictionary of Squares

Given the list of lists:

```
data = [[1, 2, 3], [4, 5], [6, 7, 8, 9]]
```

Write a program to create a dictionary where each key is a number from the lists and its value is that number squared.

Expected Output: {1: 1, 2: 4, 3: 9, 4: 16, 5: 25, 6: 36, 7: 49, 8: 64, 9: 81}

Section C: Nested Loops with Branching

These exercises combine nested loops with if-elif-else statements. Focus on understanding how conditions affect the flow within nested loop structures.

Q21. Prime Numbers in Range

Write a program that takes two numbers from the user (start and end) and displays all prime numbers in that range. Use nested loops to check for primality.

Example: start=10, end=30 → Output: 11, 13, 17, 19, 23, 29

Q22. Filter Students by Grade

Given the dictionary of students and their marks in multiple subjects:

```
students = {
    "Alice": [85, 90, 78],
    "Bob": [60, 55, 70],
    "Charlie": [92, 88, 95],
    "David": [45, 50, 40]
}
```

Write a program to display only those students whose average marks are above 70.

Q23. Number Pattern with Condition

Write a program that takes N from the user and prints a pattern where each row contains numbers from 1 to row_number, but replaces multiples of 3 with 'X'.

Example for N=5:

```
1
1 2
1 2 X
1 2 X 4
1 2 X 4 5
```

Q24. Common Elements in Multiple Lists

Given the list of lists:

```
lists = [[1, 2, 3, 4, 5], [2, 4, 6, 8], [1, 2, 3, 9, 10]]
```

Write a program to find elements that appear in ALL three lists.

Expected Output: [2]

Q25. Pascal's Triangle

Write a program that takes N from the user and prints the first N rows of Pascal's Triangle.

Example for N=5:

```
1
1 1
1 2 1
1 3 3 1
```

1 4 6 4 1

Q26. Group Products by Category and Price Range

Given the list of products:

```
products = [
    {"name": "Laptop", "category": "Electronics", "price": 50000},
    {"name": "Phone", "category": "Electronics", "price": 25000},
    {"name": "Shirt", "category": "Clothing", "price": 1500},
    {"name": "Jeans", "category": "Clothing", "price": 2500},
    {"name": "Watch", "category": "Electronics", "price": 8000}
]
```

Create a dictionary that groups products by category, but only includes products priced above 2000.

Q27. Checkerboard Pattern

Write a program that takes N from the user and prints an N×N checkerboard pattern using '0' and '1'.

Example for N=5:

```
0 1 0 1 0
1 0 1 0 1
0 1 0 1 0
1 0 1 0 1
0 1 0 1 0
```

Q28. Find Duplicate Elements

Given the list:

```
numbers = [4, 2, 7, 2, 8, 4, 9, 7, 1, 4, 3, 7]
```

Write a program to find all elements that appear more than once and count their occurrences. Store results in a dictionary.

Expected Output: {4: 3, 2: 2, 7: 3}

Q29. Spiral Number Pattern

Write a program that takes N from the user and prints numbers 1 to N² in a specific pattern where odd rows print left-to-right and even rows print right-to-left.

Example for N=3:

```
1 2 3
6 5 4
7 8 9
```

Q30. Matrix Diagonal Sum with Conditions

Given the matrix:

```
matrix = [
    [5, 3, 8, 1],
```



```
[2, 9, 4, 7],  
[6, 1, 7, 3],  
[8, 5, 2, 6]  
]
```

Write a program to calculate: (a) Sum of primary diagonal elements that are greater than 5,
(b) Sum of secondary diagonal elements that are even numbers.

— End of Exercise Set 2 —